MILITARY SPECIFICATION

SIMULATOR, LAUNCHING, ANTI-TANK, GUIDED MISSILE AND ROCKET, M22 PERFORMANCE SPECIFICATION FOR

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for design and establishes a uniform method of testing one type of Simulator, Launching, Anti-Tank, Guided Missile and Rocket, M22. The purpose of the M22 Simulator testing program is to determine performance, safety, soundness of mechanical design and resistance to environments encountered during storage, handling, and service use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of.
DOD-D-1000	- Drawing, Engineering and Associated List.
MIL-A-2550	- Ammunition and Special Weapons; General Specification For.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1370

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1168	-	Lot Numbering of Ammunition.
MIL-STD-1234	-	Pyrotechnics, Sampling, Inspection and Testing.

2.1.3 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

11749641	-	Closure Switch
11749630		Simulator, Launching Anti-Tank GM & RK, M22
9344296	-	Packing and Marking For Box, Packing Wood Wire Bound, Fiberboard, For Simulator, M22
9344297	-	Packing and Marking For Box, Fiberboard, For Simulator, M22 Launching, AT, GM & RKT

INSPECTION EQUIPMENT DRAWINGS

11749562	-	Simulator, Anti-Tank Fire

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.4 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49CFR-100-199-	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles

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(The Interstate Commerce Commission Regulations are now part of the code of federal regulations (1967) editions and revisions) available from the superintendent of documents, Government Printing Office, Washington, DC 20315. Orders for the above publications, cite 49CFR-100-199 (Latest Revision).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 Engineering design test. This specification makes provisions for engineering design testing.

3.2 Assembly. The simulator assembly shall comply with all requirements specified in Dwg. 11749630 and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Design and construction. The components of the simulator shall be designed and constructed to conform to the performance requirements specified herein.

3.3.1 Assembly of components. The components of the simulator shall consist of a Case and Closure Switch in accordance with Dwg. 11749630.

3.3.2 Contact closure switch. The output end of the cartridge shall contain an electrical contact strip (contact closure switch, refer to part no. 11749641) . This strip shall have less than 1 ohm resistance when measured as specified in 4.5.4.

3.3.3 Static and other electric hazard. The ignition system shall be designed and maintained so that the static charges and induced electric energies will not produce a hazardous device.

3.4 Firing device.

3.4.1 Performance characteristics. The simulator shall not damage the firing device (Figure 1) or interfere with repeated use.

3.4.2 Moisture content.

3.4.2.1 Charge composition. The moisture content of the charge composition at the loading station at the time of loading shall not exceed 0.10 percent, when measured as specified in para. 4.5.2.1.

3.4.2.2 Paper and paper products. The moisture content of the paper and paper products components of the simulator assembly at the loading station at the time of loading shall not exceed 6.0 percent when measured as specified in para. 4.5.2.2.

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FIGURE 1
SIMULATOR, LAUNCHING, ANTI-TANK,
MISSILE FIRE
PART NO. 11749562

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3.4.3 Performance reliability. The simulator shall meet all performance requirements at a reliability of not less than 95 percent at a confidence level of 90 percent over a temperature range of 125 ± 5°F to minus 25 ± 5°F. (See Para. 4.4.2)

3.5 Performance. The simulator shall satisfy all design requirements specified herein after subjected to the applicable environmental tests. Performance shall be measured and values recorded in accordance with 6.4 for the ballistic operation of the simulator.

3.5.1 Functioning. (Ambient temperature, 21 ± 3°C, 70 ± 5°F; Hot 51.5 ± 3°C, 125 ± 5°F; Cold minus 32 + 3°C, minus 25 ± 5°F). The assembly shall function and comply with the following requirements after conditioning at specified temperature.

3.5.1.1 Sound characteristics. The sound level intensity of the cartridge functioning at a distance of 64 + 1.0 feet (ft.) from the simulator assembly shall be 134 +6 - 7 decibles (db), when tested as specified in 4.5.3.1.

3.5.1.2 Functioning time. The time from application of nine plus or minus one (9 ± 1) volts direct current (VDC), with no current limitations, to audible report of the assembly shall not be greater than 0.110 seconds when tested as specified in 4.5.3.2.

3.5.1.3 Flash characteristics. The Simulator shall produce a visible flash or luminous plume when fired from a launch tube. This flash shall be a minimum (rein) of six (6) feet in length, two (2) feet in width in winds of not more than 8 miles per hour. Residual smoke shall be white in appearance. The simulator shall meet the performance requirements when tested per para. 4.5.3.3.

3.6 Environmental conditions.

3.6.1 Vibration. The simulators, when packaged shall not fire when subjected to vibration as specified in 4.5.8.1. After subjection to such vibration (and additional testing as indicated for First Article Tests) , the devices shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements when test fired as specified in para. 4.5.8.1.

3.6.2 Shock. The simulators, when packaged shall not fire when subjected to shocks as specified in 4.5.8.4. After subjection to such shocks (and additional testing as indicated for First Article Tests, or Quality Conformance Inspection), the devices shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements when test fired as specified in para. 4.5.8.4.

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3.6.3 Seven-foot drop. The simulators, when packaged shall not fire when subjected to drops from a height of seven feet as specified in 4.5.8.6. After being subjected to such drops (and additional testing as indicated for First Article Tests) , the devices shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements when test fired as specified in para. 4.5.8.6.

3.6.4 Loose cargo transport. The simulators, when packaged shall not fire when subjected to loose cargo transport as specified in 4.5.8.8. After being subjected to the loose cargo transport (and additional testing as indicated for First Article Tests, or Quality Conformance Inspection), the devices shall show no signs of damage, shall be safe to handle and fire, and shall meet the performance requirements when test fired as specified in para. 4.5.8.8.

3.6.5 Five-foot drop. The unpackaged simulators shall not fire when dropped from a height of five feet as specified in 4.5.8.5. After being subjected to such a drop (and additional testing as indicated for First Article Tests or Quality Conformance Inspection), the devices shall be safe to handle and fire, and shall meet the performance requirements when test fired.

3.6.6 Forty-foot drop. The simulators, when packaged shall not fire when dropped from a height of forty feet as specified in 4.5.8.7 and shall be safe to handle and dispose of.

3.6.7 Low temperature. The simulators shall be capable of withstanding low temperature conditions as specified in 4.5-8.2 and shall meet the performance requirements when test fired.

3.6.8 High temperature. The simulators shall be capable of withstanding high temperature conditions as specified in 4.5.8.3 and shall meet the performance requirements when test fired.

3.6.9 Leakage test. The simulators, when packaged shall be capable of withstanding exposure to the leakage test as specified in 4.5.6. Leakage shall constitute a failure.

3.6.10 Leak test. The Simulator shall not leak when tested as specified in 4.5.7. Leakage shall constitute a failure.

3.6.11 Jumble. The simulator shall comply with the following requirements when tested as specified in 4.5.5.

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3.6.11.1 The simulator shall not function during test.

3.6.11.2 There shall be no powder leakage.

3.6.11.3 The simulator components shall not separate during test.

3.6.12 Temperature-humidity. The packaged simulator shall withstand exposure to temperature-humidity cycle as specified in 4.5.8.9 and shall meet the performance requirements when test fired as specified in para. 4.5.8.9.

3.6.13 Temperature cycling. The simulator shall be subjected to the temperature cycling as specified in 4.5.8.10 and shall meet the performance requirements when test fired as specified in para. 4.5.8.10.

3.7 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein, the provisions of MIL-A-2550 shall apply.

4.1.1 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.3 and 3.4. If material defects, incomparability, or inferior quality is evident, and the Government deems material analysis necessary, the contractor will be requested to submit samples or specimens to the cognizant design agency for analysis and approval.

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4.2 Classification of tests. The following types of inspection shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of Table I.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in this specification.

4.3.3 Rejection.

a. If an assembly, component or test specimen fails to comply with any of the applicable requirements? the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

b. Each Environment Tests samples shall pass the accept/reject criteria for each specified environment.

c. The sum of all samples test fired (13 test for a total of 900 samples) shall pass an accept/reject criteria of 900-30-31.

d. A simulator is classified a functional failure if it fails any of the requirements of Para. 3.5.1.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63523A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Simulator, Launching Anti-Tank, GM & RK				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Simulator, Cartridge, Anti-Tank, GM RKT, M22 (Dwg. 11749630)	25		3.3.1	4.5.1
	Switch Closure (Dwg. 11749641)	25		3.3.1	4.5.1
	Closure Switch, Resistance	25		3.3.2	4.5.4
	Moisture content				
	Charge composition	25		3.4.2.1	4.5.2.1
	Paper and paper products	25		3.4.2.2	4.5.2.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63523A(AR)

PARAGRAPH	TITLE: Simulator, Launching, Anti-Tank, GM & RK, M22		SHEET 2 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY: --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Non-Sequential Test				
	Leak Test +70 ± 5°F	480		3.6.10	4.5.7
	Temperature Cycling Note 1				
	Cold, Ambient	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Cold, Ambient, Hot	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Cold, Hot	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Cold, Hot, Ambient	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Hot, Ambient	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Hot, Cold	30	1-2	3.6.13	4.5.8.10, 4.5.3
	Jumble - Ambient	30		3.6.11	4.5.5
	Forty-Foot Drop				
	Cold	1 Box		3.6.6	4.5.8.7
	Hot	1 Box		3.6.6	4.5.8.7
	Temperature - Humidity/Function 30	1 Box	1-2	3.6.12	4.5.8.9, 4.5.3
	Low Temperature Note 1	30	1-2	3.6.7	4.5.8.2,4.5.3
	High Temperature Note 1	30	1-2	3.6.8	4.5.8.3,4.5.3
	Ambient temperature Note 1	30	1-2	3.5.1	4.5.3
	Heat seal of barrier bag	30		3.2	4.5.9

NOTES:
1. Cold minus 25 ± 5°F, Ambient 70 ± 5°F, Hot +125 ± 5°F.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-S-63523A(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Simulator, Launching, AT, GM, & RKT, M22				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Sequential Test				
	Leak test +70 ± 5°F	600		3.6.10	4.5.7
	Leakage Test	4 Boxes		3.6.9	4.5.6
	Vibration - Cold	2 Boxes		3.6.1	4.5.8.1
	- Hot	2 Boxes		3.6.1	4.5.8.1
	Shock - Cold	2 Boxes		3.6.2	4.5.8.4
	- Hot	2 Boxes		3.6.2	4.5.8.4
	Seven-Foot Drop - Cold	2 Boxes		3.6.3	4.5.8.6
	- Hot	2 Boxes		3.6.3	4.5.8.6
	Loose cargo - Cold	2 Boxes		3.6.4	4.5.8.8
	- Hot	2 Boxes		3.6.4	4.5.8.8
	Leakage Test +70°F	4 Boxes		3.6.9	4.5.6
	Five foot drop - Cold-3 Boxes (Note 2)	30	1-2	3.6.5	4.5.8.5/4.5.3
	- Hot -3 Boxes (Note 2)	30	1-2	3.6.5	4.5.8.5/4.5.3
	Functioning - Ambient	540	18-19	3.5.1	4.5.3

NOTES:
2. Remove 3 boxes (9344297) from above boxes.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. See MIL-A-48078

4.4.1.2 Minimum lot size. The minimum lot size is 8,000 Simulators.

4.4.2 Examination. The examinations/tests listed herein shall be performed on inspection lots as defined in 4.4.1 of this specification.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Level II.

b. Each environment Test Samples shall pass the accept/reject criteria for each specified environment.

c. The sum of all samples test fired (5 tests for a total of 315) shall pass an accept/reject criteria of 315-7-8.

d. A Simulator is classified a functional failure if it fails any of the requirements of Para.3.5.1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63523A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Simulator, Launching AT, GM & RKT, M22				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Simulator - Physical Characteristic and Dimensions	50	0-1	3.2	4.5.1
	Closure Switch, Resistance	50	0-1	3.2.2	4.5.4
	Moisture content				
	Charge composition	50	0-1	3.4.2.1	4.5.2.1
	Paper & paper products	50	0-1	3.4.2.2	4.5.2.2

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.1	Simulator, Launching AT, GM & RKT, M22		See Below
			NEXT HIGHER ASSEMBLY --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Leak Test	315	0-1	3.6.10	4.5.7
	Non-Sequential Test Note 1				
	High Temperature	50	1-2	3.6.8	4.5.8.3,4.5.3
	Low Temperature	50	1-2	3.6.7	4.5.8.2,4.5.3
	Five Foot Drop	25		3.6.5	4.5.8.5
	Functioning - Ambient	25	1-2	3.5.1	4.5.3
	Jumble	25		3.6.11	4.5.5
	Sequential Test - Packaged +70 ± 10°F	Note 2			
	Leakage Test	75		3.6.9	4.5.6
	Shock	75		3.6.2	4.5.8.4
	Loose Cargo	75		3.6.4	4.5.8.8
	Leakage Test	75		3.6.9	4.5.6
	Functioning - Ambient	75	2-3	3.5.1	4.5.3
	Functioning - Ambient	115	2-3	3.5.1	4.5.3

NOTES: 1. Cold minus 25 ± 5°F, Ambient 70 ± 5°F, Hot +125 ± 5°F To be conditioned for six (6) hours (min.) and fired within five (5) minutes of removal from temperature chamber.

2. One half of full box of simulators (75) subjected in sequence indicated. The half of box not containing simulators (top or bottom) shall be tightly packed with dunnage of approx. configuration, wt., rigidity of replaced simulators.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.3.1 Moisture content (see Table II) - Major Defects.

TABLE II

Charge composition	(see 3.4.2.1)
Paper and paper products	(see 3.4.2.2)

The Contractor shall provide adequate controls to insure that the materials comply with the requirements. For verification, the contractor shall select and subject to test, one sample of each material from each eight (8) hours production. If a composite sample exceeds the requirement, that quantity or sub-lot material represented by the sample shall not be used in loading and packing, the remaining unloaded and unpacked material shall not be used in production, and the loaded and packed simulators shall be rejected. Test shall be performed as specified in 4.5.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Examination of product. Examination of the simulators shall be made to detect visible defects and to determine that the items meet the requirements of this specification with respect to weight, finish, and marking, and that they conform to the applicable drawing. (See 4.4.2.1)

4.5.2 Moisture content.

4.5.2.1 Charge composition. The method of determining the percentage of moisture in the flash and bang charges shall be in accordance with Method 102.2.1 of MIL-STD-1234 except that the sample shall be heated until a constant weight is obtained in lieu of four hours. Observation shall be made for compliance with applicable requirement 3.4.2.1.

4.5.2.2 Paper and paper products. The moisture content shall be determined in accordance with Method 102.1.1 of MIL-STD-1234 except that a ten (10) gram sample shall be used and the drying time shall be for two (2) hours. Observation shall be made for compliance with requirement 3.4.2.2.

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4.5.3 Functioning. (Ambient temperature, 21 ± 3°C, 70 ± 5°F, Hot, 51.5 ± 3°C, 125 ± 5°F, Cold, minus 32 ± 3°C, minus 25 ± 5°F). The simulator assemblies shall be conditioned for 12 hours min. at the specified temperature and fired within 5 minutes after removal from the conditioning chamber. The simulator shall be mounted in an approved fixture (simulating the Antitank, Missile Fire - Fig. 1) on a rigid support and fired by means of an approved power source in accordance with 3.5.1.2 (Destructive Test).

4.5.3.1 Sound characteristics. Measurement of the sound intensity shall be taken at right angle to the test fixture. The test item and sound detector shall be positioned 6 + .2 feet above ground. Observation shall be made for compliance with requirement 3.5.1.1.

4.5.3.2 Functioning time. The time sensing\reloading equipment shall be in accordance with 4.4.4. Observation shall be made for compliance with requirement of para. 3.5.1.2.

4.5.3.3 Flash characteristics. The size of the flash/plume may be determined in accordance with standard practice. These methods can include, but not be limited to, visual or photographic processes using a grid background. Observation shall be made for compliance with requirement 3.5.1.3.

4.5.4 Contact closure switch. The resistance of the closure switch shall be measured from edge to edge using any type of calibrated ohmmeter. The ohmmeter must be equipped with sharp pointed probes which can penetrate any protective coating on the contact switch. The test sample shall meet the requirement of 3.3.2. Parts tested shall not be returned to the lot. (Destructive Test)

4.5.5 Jumble. The simulator assemblies shall be tested in accordance with procedures specified in MIL-STD-331, Test 102.2, except that the duration of the test shall be a minimum of 20 minutes. The assemblies and equipment shall be observed and inspected for compliance with 3.6.11. (Destructive Test)

4.5.6 Leakage test. The test shall be performed in accordance with the procedure specified in MIL-P-116H, para. 4.4.3.1 and 4.4.3.2 or 4.4.3.6. After the test, the seal shall be examined for compliance with the applicable requirements as specified in para. 3.6.9. (Non-Destructive Test)

4.5.7 Leak test. The Simulator shall be placed in a sealed test chamber and the chamber pressurized with 3 to 5 p.s.i.g. dry air for 15 seconds. Observe pressure gage for pressure drop. Any test item that exhibits a pressure drop shall be classified defective as specified in para. 3.6.10. (Non-Destructive Test).

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4.5.8 Environmental tests. Environmental testing shall be conducted to determine compliance with the requirements of 3.6.

4.5.8.1 Vibration. This test shall be conducted in accordance with Method 514, MIL-STD-810, Test Procedure I, Test Condition 1-3.2.1, Figures 514.3-1, 514.3-2, 514.3-3 (Vertical, Transverse and Longitudinal axis) for 120 minutes (equal to 2000 miles) in each axis. Observation shall be made for compliance with requirement 3.6.1.

4.5.8.2 Low temperature. This test shall be conducted in accordance with Method 502.2, Procedure I of MIL-STD-810, except that the simulator shall be exposed to a temperature of minus 25 ± 5°F until thermal stabilization is reached and maintained for a period of 6 hours (min.). Then the simulator shall be test fired within a period of time not to exceed 5 min. Observation shall be made for compliance with requirement 3.6.7.

4.5.8.3 High temperature. This test shall be conducted in accordance with Method 501.2, procedure I of MIL-STD-810 except that the simulators shall be exposed to a temperature of 125 ± 5° until thermal stabilization is reached, and maintained at that temperature for a period of 6 hours (min.). Then the simulator shall be test fired within a period of time not to exceed 5 minutes. Observation shall be made for compliance with requirement 3.6.8.

4.5.8.4 Shock. This test shall be conducted in accordance with Method 516.3 Procedure IV of MIL-STD-810, Table 516.3-11. Use column with Largest Dimension, 91 cm and over, and Weight, under 45.4 KG (100 lbs). The 26 drops shall be divided among four boxes (two boxes at minus 25 + 5°F, two boxes at +125 ± 5°F. The packaged simulators shall be observed for compliance with requirement 3.6.2.

4.5.8.5 Five-foot drop test. The simulators shall be fired after a drop from 5 feet on to a 2-inch thick steel plate embedded in concrete. During First Article Testing, each item shall be dropped once for each orientation listed below at hot temperature (six per orientation, total of 30 items); followed by the same at cold temperature (six per orientation, total of 30 items). During Quality Conformance Inspection, one (1) item shall be dropped once for each orientation (total 5 items per orientation). Observation shall be made for compliance with requirement 3.6.5.

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a. Bottom down.
b. Top down.
c. Side down.
d. Bottom down at 45° angle.
e. Top down at 45° angle.

4.5.8.6 Seven foot drop. The packaged simulators shall be dropped from a height of 7 feet (measured to the bottom of the package) on to a 2-inch thick steel plate embedded in concrete in the orientations below:

a. One drop on side and one drop on end; at hot temperature.
b. One drop on top and one on bottom at 45° angle; at hot temperature.
One drop with bottom down and one drop on top edge at 45° angle; at cold temperature.
d. One drop on side (opposite (a) side) and one drop on one end (opposite (a) side) ; at cold temperature.

The packaged simulators shall be observed for compliance with requirement 3.6.3.

4.5.8.7 Forty-foot drop test. This test shall be conducted in accordance with Test 103 of MIL-STD-331. The packaged simulators shall be dropped in the following positions: One drop with bottom down at minus 25°F, one drop with end down at 125°F. A new simulator package shall be used for each drop. The simulator shall be observed for compliance with requirement 3.6.6 (Destructive Test).

4.5.8.8 Loose cargo. This test shall be conducted in accordance with Method 514.3 of MIL-STD-810, Test Procedure II, Test Conditions I-3.2.3. (One box positioned on its base and one box positioned on its end, at hot and cold temperature for first article only) . The simulators shall be observed for compliance with requirement 3.6.4.

4.5.8.9 Temperature-humidity. The packaged simulators shall be subjected to a 14 day temperature humidity test according to MIL-STD-331, Test 105.1. Observation shall be made for compliance with requirement 3.6.12.

4.5.8.10 Temperature-cycling. The packaged items shall be conditioned Cold (minus 25 ± 5°F), Ambient (±70°F), and Hot (+125°F)r as specified in MIL-STD-810, Method 503.2, Procedure I, Table 503.2-1. Hot Dry Ambient Temperature item response will be 125 ± 5°F. Observation shall be made for compliance with requirement 3.6.13.

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4.5.9 Heat seal of barrier bag. (See Dwg. 9344297). The test shall be performed in accordance with the procedure specified in MIL-P-116H para. 4.4.4.3. After the test, the seal shall be examined for compliance with the applicable requirements. (Non-Destructive Test)

5. PACKAGING

5.1 Cleaning. The simulator shall be cleaned in accordance with Process C-1 of MIL-P-116, as required.

5.2 Drying. The simulator shall be dried in accordance with MIL-P-116, as required.

5.3 Preservation. Not required.

5.4 Packaging requirements. Packaging shall be as specified on Dwgs. 9344296 and 9344297.

5.5 Packing and marking. Packing and marking shall be as specified on Dwgs. 9344296 and 9344297.

5.6 Department of transportation. DOT authorization and classification to ship shall be obtained by the supplier. A COPY Of the DOT authorization shall be furnished to the Contracting Officer prior to delivery of any assemblies/items/hazardous materials.

5.6.1 DOT authorization/harzard classification. Certification from the supplier shall be required to assure no significant change in the item, materials, packaging, etc. which would affect the DOT authorization/hazard classification. Should any change in the DOT marking shown on drawing 9344296 be required by the DOT IAW Title 49 CFR, notification will be provided through the contracting officer to the design agency.

6. NOTES

6.1 Intended use. The simulator is a device utilizing energy produced by explosives, propellants, or pyrotechnics or combination thereof to produce a flash, bang and smoke signature. This is a training device used with the Hoffmann Device (Figure 1) on several armored vehicles.

6.2 Hazard Notes. The provisions as cited in the following paragraphs are applicable to this specification.

6.2.1 Hazardous materials. Hazardous materials/components will be identified as they are selected and submitted to the Procuring Contracting Officer for approval. Only energetic materials acceptable for DOD use will be utilized.

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6.2.2 Hazard analyses. Hazard analyses and risk assessments will be performed and tailored to the magnitude of the developmental changes required to produce an acceptable product. A hazard analysis and risk assessment of the production facilities shall be performed to assure a safe operation in accordance with DI-H-7048. These analyses shall be made available to the government for review.

6.2.3 Explosive ordnance disposal. Explosive Ordnance Disposal Render Safe Procedures and tools will be formulated as early as possible and will be kept current with design changes of the simulator as they are instituted. These procedures and tools shall be made available to the government for approval as early as possible to insure applicable tools, equipment and publications will be available to Army EOD units 30 days before the deployment date of the explosive ordnance.

6.2.4 Demilitarization. Demilitarization/disposal procedures will be formulated and kept current with design changes as they are instituted. These procedures shall be made available to the government.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAT-I (D), Dover, NJ 07801-5001.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Retirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.5	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724
4.4.4	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-10054
6.2.2	Hazard Analyses	DI-H-7048

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Contractors in connection with specific acquisition functions should obtain copies from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.5 Contractor's design requirements and specifications.

6.5.1 Inert and explosive materials. All inert and explosive materials used in the Simulator shall be referenced to approved Government specifications if such exist. Proprietary materials should not be used unless specifically approved in writing by the contract officer. At the direction of the contract officer, the contractor should submit simulator drawings, in sufficient detail as to parts, special processes, and techniques to permit the preparation of documentation in accordance with 6.5.2.

6.5.2 Drawings. Drawings should be in accordance with DOD-D-1000.

6.6 Definition of packaging and packing levels.

6.6.1 Level A. Full military protection for unrestricted (OCONUS) shipment and storage.

6.6.2 Level B. Limited military protection for limited storage and (OCONUS) shipment.

6.6.3 Level C. For interplant (CONUS) shipment and very limited storage.

6.7 Safety. This specification contains requirements which may necessitate the handling of hazardous materials and result in the manufacture of a hazardous item. It is emphasized that all applicable safety rules, procedures, and regulatory requirements be strictly followed during the handling, processing, and testing of the completed item and related hazardous materials.

6.7.1 Safety certification. Certification from the supplier and/or first article testing shall be required to assure no significant change in the item, materials, packaging, etc. which would affect the DOT authorization/hazard classification.

6.8 Noise protection. Noise protection is advisable to prevent ear damage whenever safe distance of 96 meters is not adhered to.

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6.9 Changes from Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-63523A	SIMULATOR, LAUNCHING, ANTI-TANK, GM FOR ROCKET M22

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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